

LTL

In LTL we have a Boolean state or world in which a number of Atomic Propositions (AP) are True or False. $\rightarrow$ Each AP is just a Boolean variable

We consider discrete linear time - an infinite sequence of states $A_0 A_1 A_2 \dots$, where A_0 at time 0.

$\Rightarrow$ Any propositional formula in the AP refers to A_0 .

We have these operations (as well as $\wedge, \vee, \neg$), in terms of a propositional formula ϕ :

$\bigcirc \phi$: ϕ holds in the next state

$\Diamond \phi$: ϕ holds eventually at some point in the future.

$\Box \phi$: ϕ always holds at all future states.

$\phi_1 U \phi_2$: ϕ_1 must always hold until ϕ_2 occurs $\rightarrow \phi_2$ must eventually occur.

$\Rightarrow$ At the point ϕ_2 occurs and afterwards, ϕ_1 can be either true or false.

All LTL operations can be written using the following operations:

Definition. A formula in LTL is defined by the following grammar (in which brackets are omitted)

$$\phi := \text{true} \mid a \mid \phi_1 \wedge \phi_2 \mid \neg \phi \mid \bigcirc \phi \mid \phi_1 U \phi_2,$$

where $a \in AP$ and ϕ_1, ϕ_2 are LTL formulae.

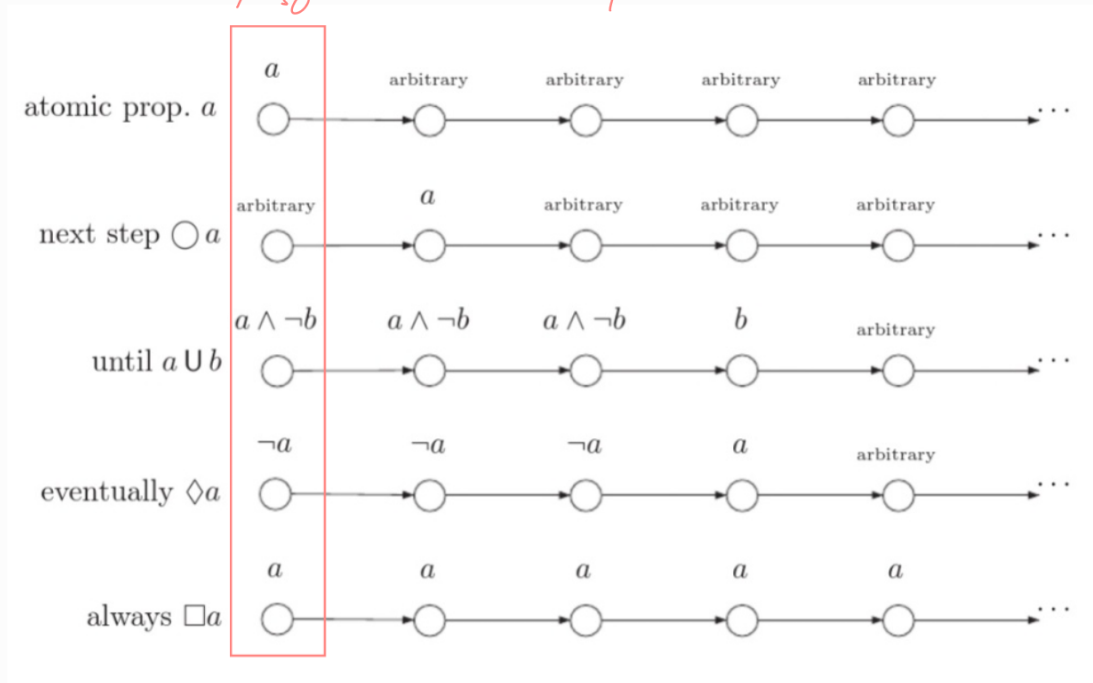
$$\Diamond a \equiv \text{true} U a$$

$$\Box a \equiv \neg \Diamond \neg a$$

U enforces that the second formula must eventually hold.

$$\neg(\Diamond(\neg a)) \rightarrow \neg(\text{eventually } \neg a \text{ occurs})$$

A_0 - the state all formulae are talking about.



The set of atomic propositions AP is a set of Boolean variables whose values can be either True or False.

Each world, or state, is represented by the subset of AP that are True at this state. This means that each world is a letter from the alphabet 2^{AP} .

Note: 2^{AP} is the set of all possible subsets of AP , i.e. the set of all possible worlds.

A word (or trace) σ is an infinite sequence of worlds $A_0 A_1 A_2 \dots$, i.e. $\sigma \in (2^{AP})^\omega$

Formally, the satisfaction relation, $\sigma \models \varphi$, is defined as:

(Notes: - $\sigma \models \varphi$ means the word σ satisfies the formula φ .
- we only consider whether formulae are satisfied at A_0 .)

$\sigma \models \text{true}$

$\sigma \models a$ iff $a \in A_0$

$\sigma \models \varphi_1 \wedge \varphi_2$ iff $\sigma \models \varphi_1$ and $\sigma \models \varphi_2$

$\sigma \models \neg \varphi$ iff $\sigma \not\models \varphi$

$\sigma \models \bigcirc \varphi$ iff $A_1 \dots \models \varphi$

$\sigma \models \varphi_1 \cup \varphi_2$ iff there is $i \geq 0$ s.t. $A_i \dots \models \varphi_2$ and $A_j \dots \models \varphi_1$ for all $0 \leq j < i$.

$\text{Words}(\varphi)$ is the set of all possible words, or traces, that satisfy φ .

A Transition System is a model that describes how a system changes over time, containing states, transitions and labels. A TS is defined as a 5-tuple $(S, I, \rightarrow, AP, L)$, where:

- S is a finite set of states
- I is a set of initial states, $I \subseteq S$
- $\rightarrow$ is a transition relation, defined such that every state $S_1 \in S$ has an outgoing transition to some other state $S_2 \in S$ so that the execution is infinite, $\rightarrow \subseteq S \times S$
- AP is a finite set of atomic propositions
- L is a labelling function that labels each state with the subset of AP that are true at that state, $L: S \rightarrow 2^{AP}$

(Note: if transitions are labelled by a finite set of Actions, Act , then $\rightarrow \subseteq S \times Act \times S$)

A run on a TS is an infinite sequence of states

$$S_0 \rightarrow S_1 \rightarrow S_2 \rightarrow \dots$$

Such that $S_0 \in I$.

This run produces an infinite trace, $\sigma \in (\Sigma^{AP})^\omega = L(S_0) L(S_1) \dots$, i.e. the trace is the word represented by the labels of the states traversed in the run.

Note: A path on a TS is the same as a run, but S_0 does not have to be in I .

The set of all possible traces of the TS is $\text{Traces}(TS)$

A TS satisfies a formula φ if $\text{Traces}(TS) \subseteq \text{Words}(\varphi)$ — i.e. every possible trace of the TS satisfies φ .

Note: If some traces satisfy φ and some satisfy $\neg\varphi$ then $TS \not\models \varphi$ and $TS \not\models \neg\varphi$.